AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 14 · STUDENT LESSON

Statistical tests and assumptions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

State the null. Enumerate the signs. Count inclusive extremes.

Calculate an exact paired sign-flip p-value while preserving pairing, arithmetic scale and non-rejection limits.

Keep one habit: Paired-null contract → all-eight-assignment ledger → native paired-time null chart → tie/scale probes → rational enumeration → independent rescaled differences.

MAIN LESSON · SECTION 1

Compare a statistic with an explicit null distribution

> Keep valid pairing, sign assumptions, tail rule and decision threshold visible.

Learning goals

I can explain the **mechanism** and terminology of statistical tests and assumptions.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve effect magnitude, practical thresholds and sampling assumptions

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Effect size and practical importance. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain statistical tests and assumptions using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of statistical tests and assumptions

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Define the null model and tail rule before calculating p

> An exact arithmetic result can still be invalid if the study assumptions do not hold.

Begin with the null-model question

A statistical test asks how extreme the observed statistic is under a specified null model. Here three valid independent paired differences have magnitudes one, one and one. Under the assumed symmetric zero-effect null, each sign can be plus or minus independently with equal probability. List all eight assignments and count absolute means at least as large as the observed one.

This tail probability is not the probability that the null is true, and the enumeration does not create more observed pairs.

Define a statistical test and p-value

A statistical test compares observed **evidence** with a specified **null model**. A **p-value** is not the probability that the null hypothesis is true.

Complete pairing, sign and decision contract

Exact-test item	Complete constructed sign-flip contract
Main observed differences	1, 1, 1 from three independent paired units
Null assumptions	Valid pairing; independent equiprobable signs conditional on fixed magnitudes under the symmetric zero-effect null
Not sufficient by itself	A statement that the mean is zero does not establish the required sign exchangeability
Null enumeration	All eight sign assignments; each has probability 1/8
Two-sided statistic	Absolute mean; with fixed n, absolute sum gives the same extremeness ordering
Inclusive tail rule	Count assignments at least as extreme, including ties
Decision rule	Predeclared alpha 0.05; reject only if exact p ≤ alpha
Arithmetic	Exact integers/Fraction values in complete code; no arbitrary absolute tolerance
Scope	Conditional toy calculation; pairing, independence, symmetry and actual randomization are assumed, not verified by code

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace all eight sign assignments and count both extremes

> The all-plus and all-minus arrangements both count as equally extreme.

Original conditional sign-flip procedure

Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all 2^n sign assignments, holding magnitudes fixed. Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.

Complete signed-sum/absolute-mean ledger

Sign assignment for magnitudes 1, 1, 1	Signed sum	Absolute mean	At least observed 1?
− − −	−3	1	Yes
− − +	−1	1/3	No
− + −	−1	1/3	No
− + +	1	1/3	No
+ − −	−1	1/3	No
+ − +	1	1/3	No

Sign assignment for magnitudes 1, 1, 1	Signed sum	Absolute mean	At least observed 1?
+ + −	1	1/3	No
+ + +	3	1	Yes
Inclusive exact tail	Two of eight assignments	p=2/8=0.25	Above alpha 0.05: do not reject

State model assumptions and coarse test resolution

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Statistical tests and assumptions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05.	Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all 2^n sign assignments, holding magnitudes fixed. Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.	Only all-plus and all-minus have absolute mean 1; others have 1/3. Exact p=2/8=.25. At alpha .05 do not reject; this does not prove zero effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict what happens when legitimate task pairing is removed

> Unrelated measurements do not inherit the paired null model.

Current task and response

- Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict broken pairing

- Change: Break the task pairing and use unrelated A and B measurements.

▼ Check after predicting

Prediction: This paired sign-flip calculation is no longer justified by the stated design.

Reason: Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Read observed paired differences against the conditional null distribution

> Keep the picture's absolute-sum calculation consistent with the main absolute-mean ordering.

Main one/one/one differences and checked result

Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05.

Only all-plus and all-minus have absolute mean 1; others have 1/3. Exact p=2/8=.25. At alpha .05 do not reject; this does not prove zero effect.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate paired-time inputs and model scope

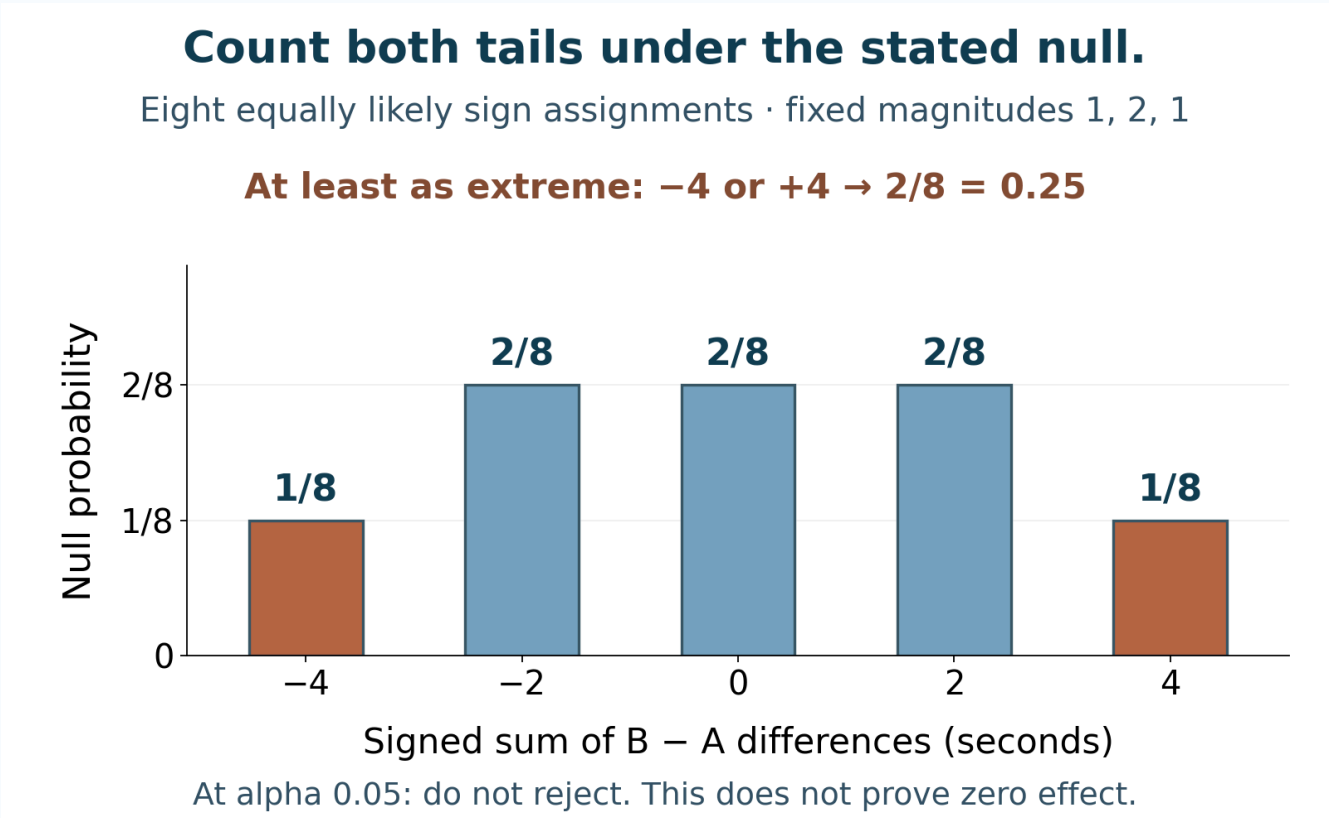
A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

A times [5,8,4] seconds; B [4,6,3]. Differences B–A are [-1,-2,-1]. Use a two-sided statistic: absolute sum of differences. Assume the stated sign-symmetry null is appropriate.

The picture uses separate paired times A=[5, 8, 4] and B=[4, 6, 3] seconds. B–A differences are–1, –2, –1, with sum –4 and mean –4/3 seconds. Fix magnitudes 1, 2, 1 and enumerate eight equally likely sign arrangements under the declared null. Signed sums–4, –2, 0, 2, 4 occur 1, 2, 2, 2, 1 times. Only–4 and+4 have absolute sum at least 4, so p=2/8=0.25.

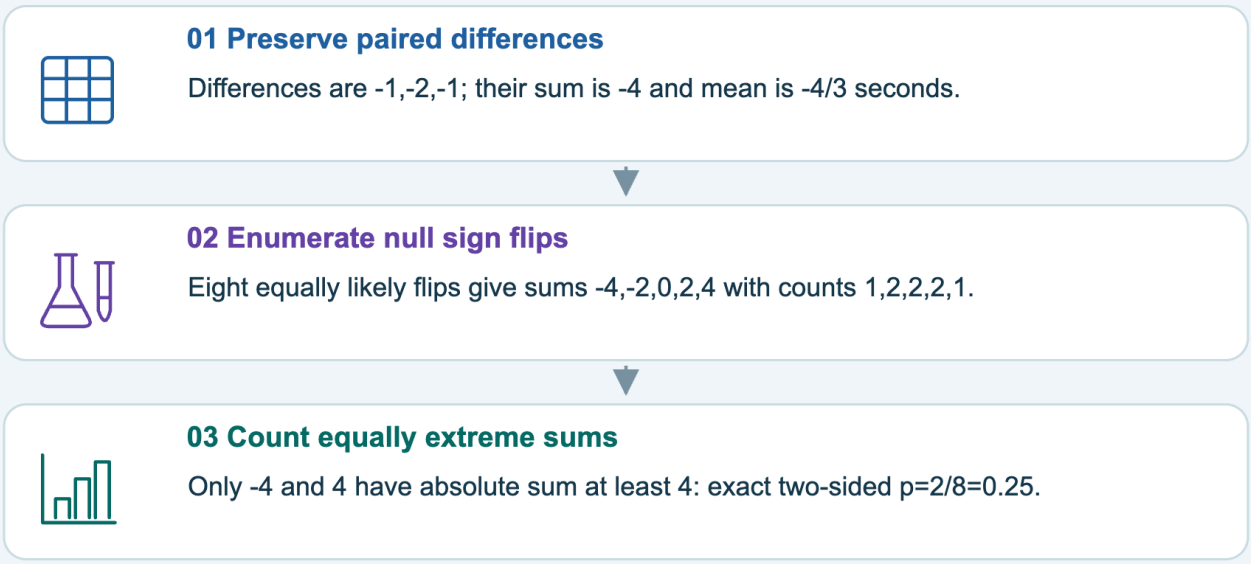
Do not reject at 0.05; neither zero effect nor practical unimportance is proved. If the A/B measurements are unrelated rather than valid pairs, this paired analysis is unjustified. Exact arithmetic cannot repair the study design.

See both null tails in the native chart



AI ACADEMY / AI-7/14 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Preserve paired differences	Differences are -1,-2,-1; their sum is -4 and mean is -4/3 seconds.
2. Enumerate null sign flips	Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1.
3. Count equally extreme sums	Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A test compares an observed statistic with outcomes under a specified **null model**. Here the sign-flip assumptions make eight arrangements equally likely.

At a predeclared 0.05 threshold, p=0.25 does not reject the null; it does not prove no difference.

Complete paired-time/statistic/tail ledger

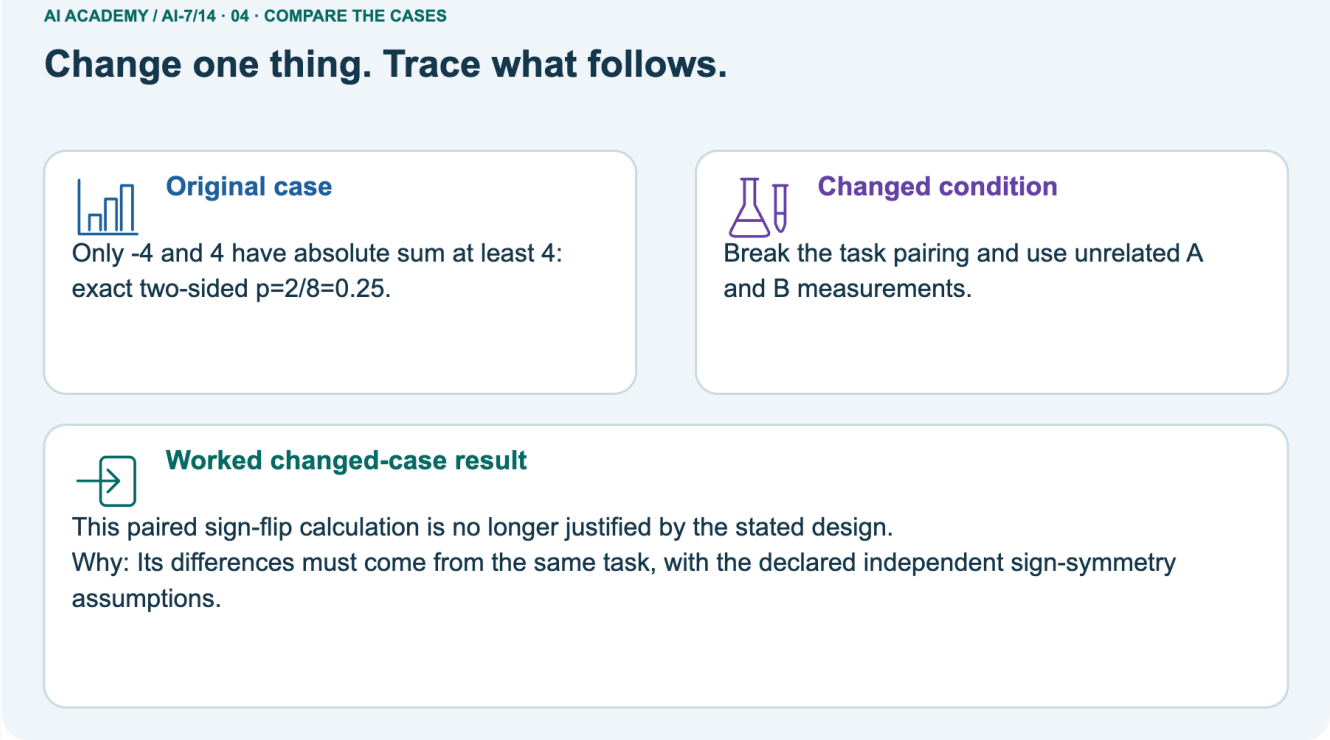
Separate picture quantity	Complete supplied calculation
Paired B−A differences	4−5=−1; 6−8=−2; 3−4=−1 seconds
Observed sum / mean	−4 seconds / −4/3 seconds
Fixed null magnitudes	1, 2, 1
Signed-sum counts	−4:1; −2:2; 0:2; 2:2; 4:1, total 8

Separate picture quantity	Complete supplied calculation
Inclusive extreme arrangements	Absolute sum≥4:−4 and+4, one each
Exact two-sided p	2/8=0.25
Decision at alpha 0.05	Do not reject under this model
What is not proved	Null truth, zero difference, practical equivalence or valid pairing

Predict replacing pairs with unrelated observations

- Change one thing: Break the task pairing and use unrelated A and B measurements. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain invalid assumptions and non-rejection meaning

- Three pairs provide coarse **evidence**. The **p-value** is not the probability that the null is true; invalid independence or symmetry assumptions invalidate this test.

- Read the labelled picture: Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.

▼ Earlier wording, alternatives and picture transcripts

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from itertools import product
from fractions import Fraction

def sign_flip_p(differences):
    """Two-sided exact test under independent symmetric sign exchangeability.

    Tiny n only; hold absolute magnitudes fixed and include equal extremes.
    A p-value is not a posterior probability that the null is true.
    """
    if not differences or len(differences) > 16:
        raise ValueError('require 1–16 paired units')
    differences = [Fraction(d) for d in differences]
    threshold = abs(sum(differences))
    count = 0
    for signs in product((-1, 1), repeat=len(differences)):
        simulated = abs(sum((s * abs(d) for (s, d) in zip(signs, differences))))
        if simulated >= threshold:
            count += 1
    return count / 2 ** len(differences)

print(sign_flip_p([1, 1, 1]))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Statistical tests and assumptions

The facts we will use

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Three independent paired task times

	A seconds	B seconds	B-A seconds
Task 1	5	4	-1
Task 2	8	6	-2
Task 3	4	3	-1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Statistical tests and assumptions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preserve paired differences

Differences are -1,-2,-1; their sum is -4 and mean is -4/3 seconds.



2. Enumerate null sign flips

Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1.



3. Count equally extreme sums

Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

CHANGE ONE THING

Break the task pairing and use unrelated A and B measurements.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit the null model, inclusive tail and meaning of non-rejection

> A p-value is not an effect size, posterior null probability or practical decision.

Core statistical-test qualification and failure analysis

The test requires its sign-**exchangeability** and independence assumptions; do not apply it blindly to clustered or asymmetric differences.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete resolution, scale and assumption probes

Test/assumption probe	Expected interpretation
Observed statistic equals a null arrangement	Include ties in at-least-as-extreme count
Only favorable-sign extreme counted	Wrong for this declared two-sided absolute-statistic rule
Exact p = 0.25 at alpha 0.05	Do not reject; not evidence that null is true with probability 0.25
No rejection	Does not establish zero effect or practical equivalence
Three nonzero paired magnitudes in this sign-flip setup	Smallest two-sided tail is 2/8; magnitude alone cannot reach 0.05
Any small-n test in general	Do not generalize this specific resolution limit to all statistical methods
Scale all differences by the same positive constant	Extremeness ordering and exact p should stay unchanged
Absolute numeric epsilon overwhelms tiny values	Can corrupt ties/tail counts; use exact arithmetic here

Test/assumption probe	Expected interpretation
Pairing or sign assumptions invalid	The numerical p does not validate the design

Guided repair and explanation

- A report omits this qualification: The test requires its sign-**exchangeability** and independence assumptions; do not apply it blindly to clustered or asymmetric differences.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise naming assumptions before counting null arrangements

> Report statistic, tail count, denominator and decision separately.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example “Statistical tests and assumptions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Statistical tests and assumptions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the selected doubled-magnitude paired differences

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run a complete exact paired test with scale and tie checks

> Inspect the full null enumeration instead of accepting a p-value without its assumptions.

Current three-pair exact-test investigation

Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05. Show the intermediate steps and give the conclusion with its scope.

Runnable exact-rational null enumeration

```
"""Exact tiny-n conditional sign-flip test, under explicitly assumed valid
pairing/null."""
from itertools import product
from fractions import Fraction
from collections import Counter

def exact_rational(value):
 if type(value) is int or type(value) is Fraction:return Fraction(value)
 raise ValueError('use exact integers or Fraction values, not Boolean/inexact inputs')

def sign_flip_test(differences,alpha=Fraction(1,20)):
 """Assume independent paired units and conditionally independent equiprobable signs.
 Hold magnitudes fixed under the symmetric zero-effect null. Design assumptions
 cannot be verified from these numbers. Absolute-mean ties count inclusively.
 """
 if type(differences) not in (list,tuple) or not 1<=len(differences)<=16:
  raise ValueError('require 1 to 16 exact paired differences')
 values=[exact_rational(d) for d in differences];level=exact_rational(alpha)
 if not 0<level<1:raise ValueError('alpha must be between zero and one')
 n=len(values);observed_sum=sum(values);observed_absolute_mean=abs(observed_sum)/n
 rows=[]
 for signs in product((-1,1),repeat=n):
  signed=tuple(s*abs(d) for s,d in zip(signs,values));total=sum(signed)
  statistic=abs(total)/n

rows.append({'signs':signs,'signed_differences':signed,'sum':total,'absolute_mean':statistic,
               'at_least_as_extreme':statistic>=observed_absolute_mean})
 extreme=sum(row['at_least_as_extreme'] for row in rows)
 p=Fraction(extreme,len(rows))
 return {'observed_sum':observed_sum,'observed_absolute_mean':observed_absolute_mean,
```

```
        'rows':rows,'extreme':extreme,'assignments':len(rows),'p':p,'alpha':level,
        'decision':'reject' if p<=level else 'do_not_reject'}
main=sign_flip_test([1,1,1])
assert main['observed_absolute_mean']==1 and main['extreme']==2 and main['assignments']==8
and main['p']==Fraction(1,4)
assert main['decision']=='do_not_reject'
# Separate picture: paired B-A times [-1,-2,-1], absolute sum ordering equivalent for
fixed n.
picture=sign_flip_test([-1,-2,-1])
sum_counts=Counter(row['sum'] for row in picture['rows'])
assert sum_counts=={-4:1,-2:2,0:2,2:2,4:1} and picture['p']==Fraction(1,4)
# Scale invariance and inclusive ties require no arbitrary absolute epsilon.
scaled=sign_flip_test([Fraction(1,10**13)]*3)
assert scaled['p']==main['p'] and sign_flip_test([0,0,0])['p']==1
print('Main exact assignments:',
[(r['signs'],str(r['sum']),str(r['absolute_mean']),r['at_least_as_extreme']) for r in
main['rows']])
print('Main exact two-sided p:',main['p'],'decision at alpha 0.05:',main['decision'])
print('Picture signed-sum null counts:',{str(k):v for k,v in
sorted(sum_counts.items())},'p:',picture['p'])
print('Exact tiny-scale check:',scaled['p'],'ties included; enumeration adds no new
observed units.')
print('The p-value is a null-model tail probability, not the probability the null is true.
Do not reject does not prove zero effect or practical unimportance.')
print('No actual random assignment, model experiment or significance claim for
unpaired/correlated data is supplied. Exact arithmetic cannot verify pairing,
independence, symmetry or exchangeability.')
```

Expected p, numerical correction and study limits

1. Predict the eight main absolute means and the two extreme assignments before running or tracing. The complete exact-rational code gives p=1/4 for the main and separate picture examples, without random resampling. A tiny positive scaling check preserves the same p, and all-zero differences give p=1 with inclusive ties. These checks validate arithmetic, not real pairing or sign symmetry.

   The earlier compact displayed code is also corrected to rational arithmetic; a fixed absolute epsilon previously gave the wrong result at tiny scales. The original lab index is retained for draft compatibility. Do not treat the eight enumerated null arrangements as eight new observations or a real randomized study.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

```
from itertools import product
from fractions import Fraction

def sign_flip_p(differences):
    """Two-sided exact test under independent symmetric sign exchangeability.

    Tiny n only; hold absolute magnitudes fixed and include equal extremes.
    A p-value is not a posterior probability that the null is true.
    """
    if not differences or len(differences) > 16:
        raise ValueError('require 1–16 paired units')
    differences = [Fraction(d) for d in differences]
    threshold = abs(sum(differences))
    count = 0
    for signs in product((-1, 1), repeat=len(differences)):
        simulated = abs(sum((s * abs(d) for (s, d) in zip(signs, differences))))
        if simulated >= threshold:
```

```
            count += 1
    return count / 2 ** len(differences)

print(sign_flip_p([1, 1, 1]))
```

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: doubled differences under the same conditional null

> Enumerate all eight assignments and interpret the exact two-sided result.

Complete selected independent task

Three independent paired differences are 2, 2, 2. Assume a symmetric zero-effect null under which each sign flips independently with equal probability. Enumerate all eight assignments. The statistic is absolute mean; include ties as equally extreme. Compute the two-sided exact p-value and interpret at alpha 0.05.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Three independent paired differences are 2, 2, 2. Assume a symmetric zero-effect null under which each sign flips independently with equal probability. Enumerate all eight assignments. The statistic is absolute mean; include ties as equally extreme. Compute the two-sided exact p-value and interpret at alpha 0.05.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save paired units, null assumptions, statistic and complete tail rule

> Keep practical magnitude and inferential decision as different records.

Project task and artifact

Add a labelled statistical tests and assumptions **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of statistical tests and assumptions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local enumeration or paper trace and limitations

For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the p-value as a conditional tail probability

> Non-rejection cannot prove the null or fix an invalid experiment.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of statistical tests and assumptions

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Null model	The probability model used for a test comparison
P-value	The null-model chance of an outcome at least as extreme under the defined statistic
Exchangeability	An assumption permitting the declared rearrangements

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: The test requires its sign-**exchangeability** and independence assumptions; do not apply it blindly to clustered or asymmetric differences.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Statistical tests and assumptions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05.	Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all 2^n sign assignments, holding magnitudes fixed. Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.	Only all-plus and all-minus have absolute mean 1; others have 1/3. Exact p=2/8=.25. At alpha .05 do not reject; this does not prove zero effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

By the end, I can	How I will show it
Explain the mechanism and terminology of statistical tests and assumptions	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference’s sign to flip independently with equal probability.
- Which labelled step matches this detail? Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.
- 1. Enumerate null sign flips
- 2. Count equally extreme sums
- 3. Preserve paired differences
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Count equally extreme sums
Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

A test compares an observed statistic with outcomes under a specified **null model**. Here the sign-flip assumptions make eight arrangements equally likely.

At a predeclared 0.05 threshold, p=0.25 does not reject the null; it does not prove no difference.

Changed-case question and optional hint

- Changed-case practice: Break the task pairing and use unrelated A and B measurements. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This paired sign-flip calculation is no longer justified by the stated design.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: This paired sign-flip calculation is no longer justified by the stated design.

Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

Repair and retain the record

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

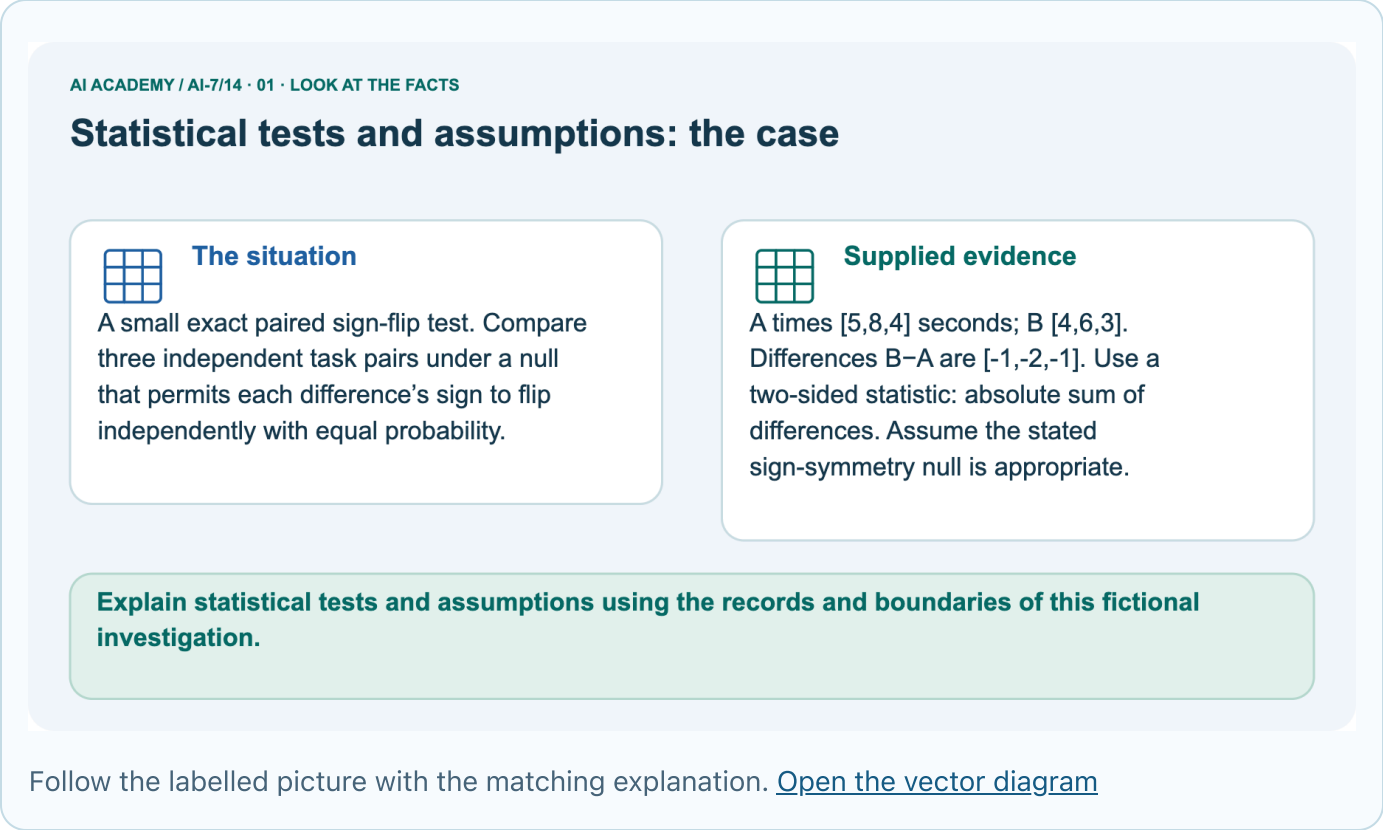
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Statistical tests and assumptions: the case



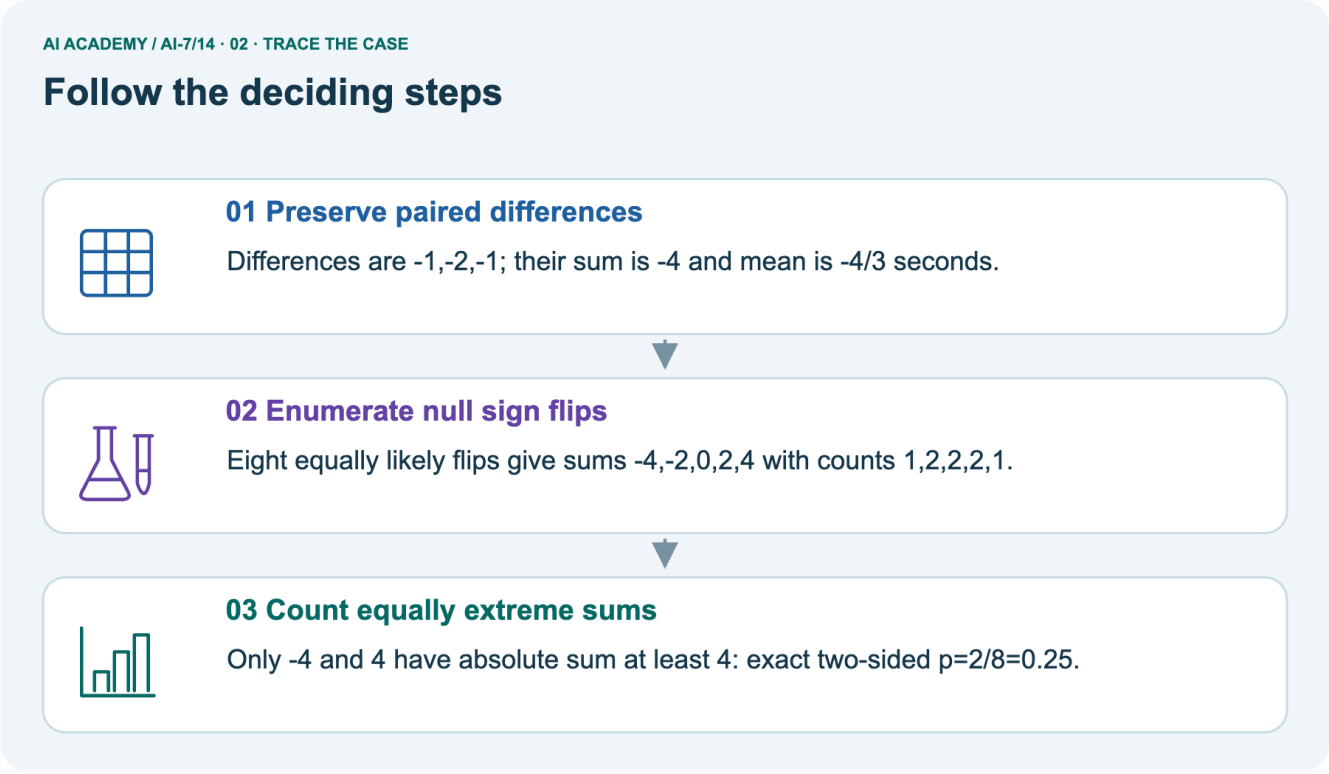
Follow the labelled picture with the matching explanation. Open the vector diagram

A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability. Supplied facts: A times [5,8,4] seconds; B [4,6,3].

Differences B–A are [-1,-2,-1]. Use a two-sided statistic: absolute sum of differences. Assume the stated sign-symmetry null is appropriate.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

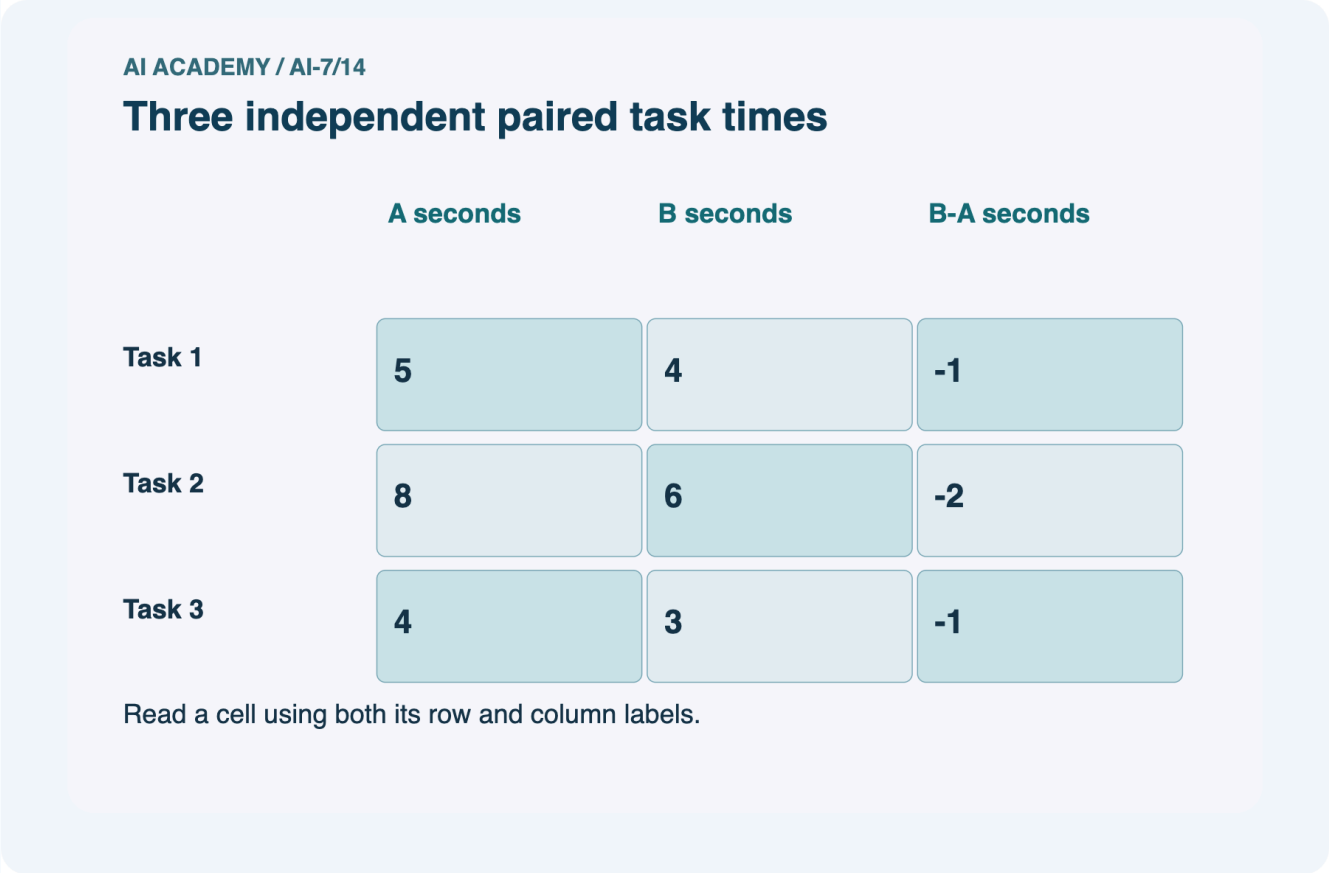


Follow the labelled picture with the matching explanation. Open the vector diagram

Preserve paired differences: Differences are -1,-2,-1; their sum is -4 and mean is -4/3 seconds. → Enumerate null sign flips: Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1. → Count equally extreme sums: Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

Point to each numbered stage and explain which recorded **input** it uses.

03. Three independent paired task times

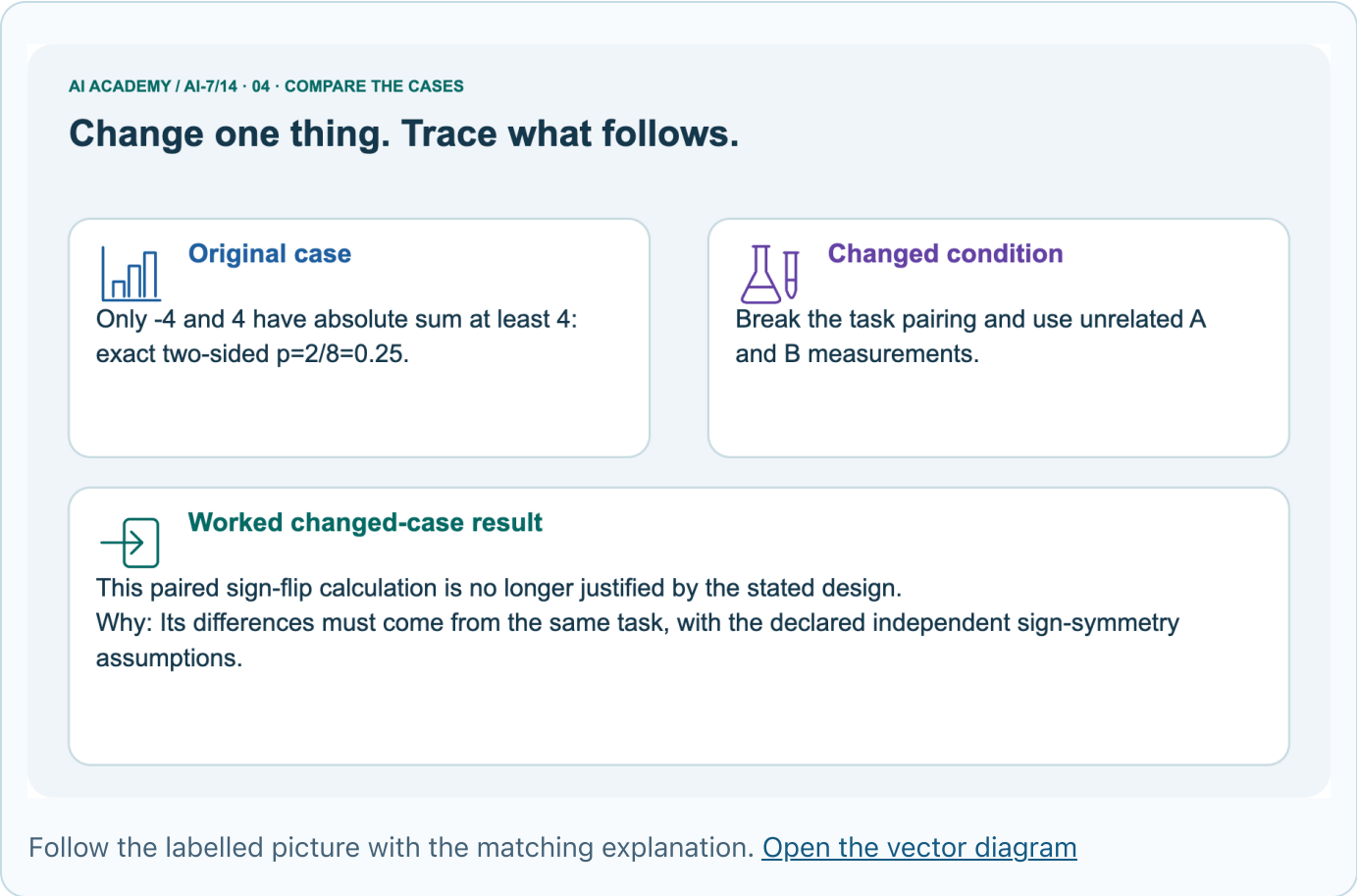


Follow the labelled picture with the matching explanation. Open the vector diagram

Three independent paired task times. Task 1: A seconds = 5, B seconds = 4, B–A seconds = -1. Task 2: A seconds = 8, B seconds = 6, B–A seconds = -2. Task 3: A seconds = 4, B seconds = 3, B–A seconds = -1.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



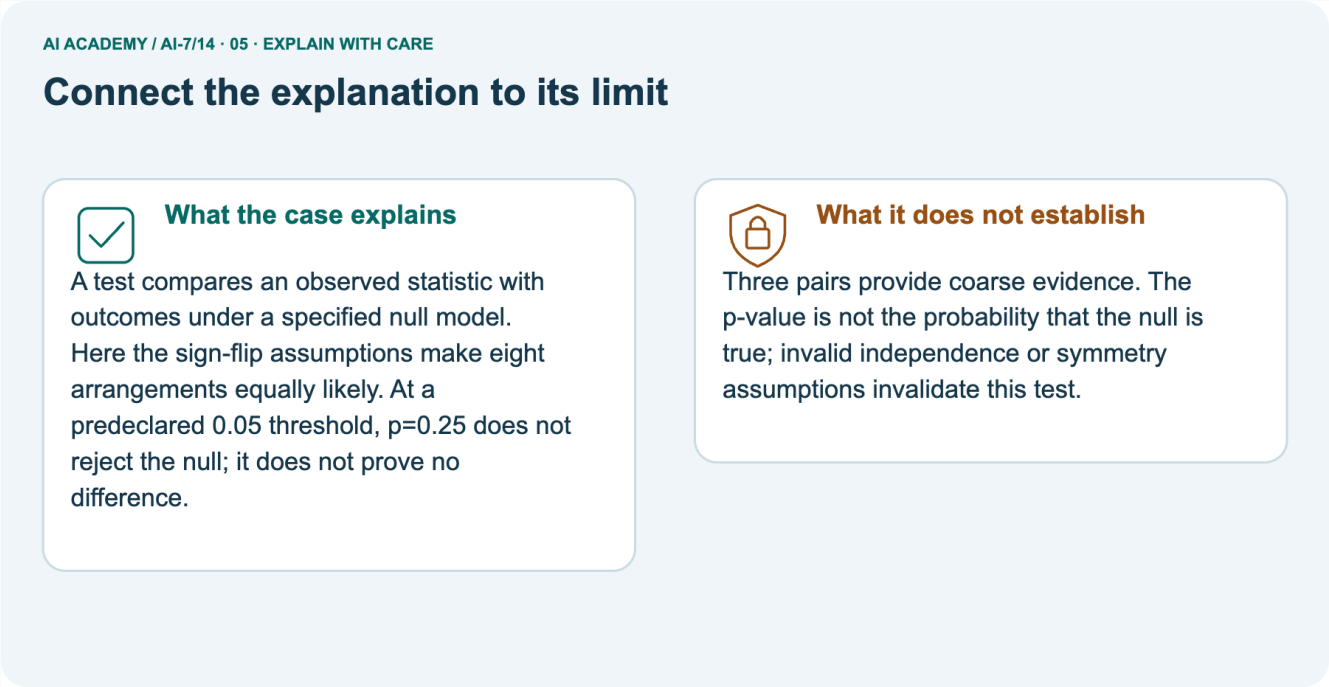
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.
Changed condition: Break the task pairing and use unrelated A and B measurements.

Expected result: This paired sign-flip calculation is no longer justified by the stated design.
Reason: Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

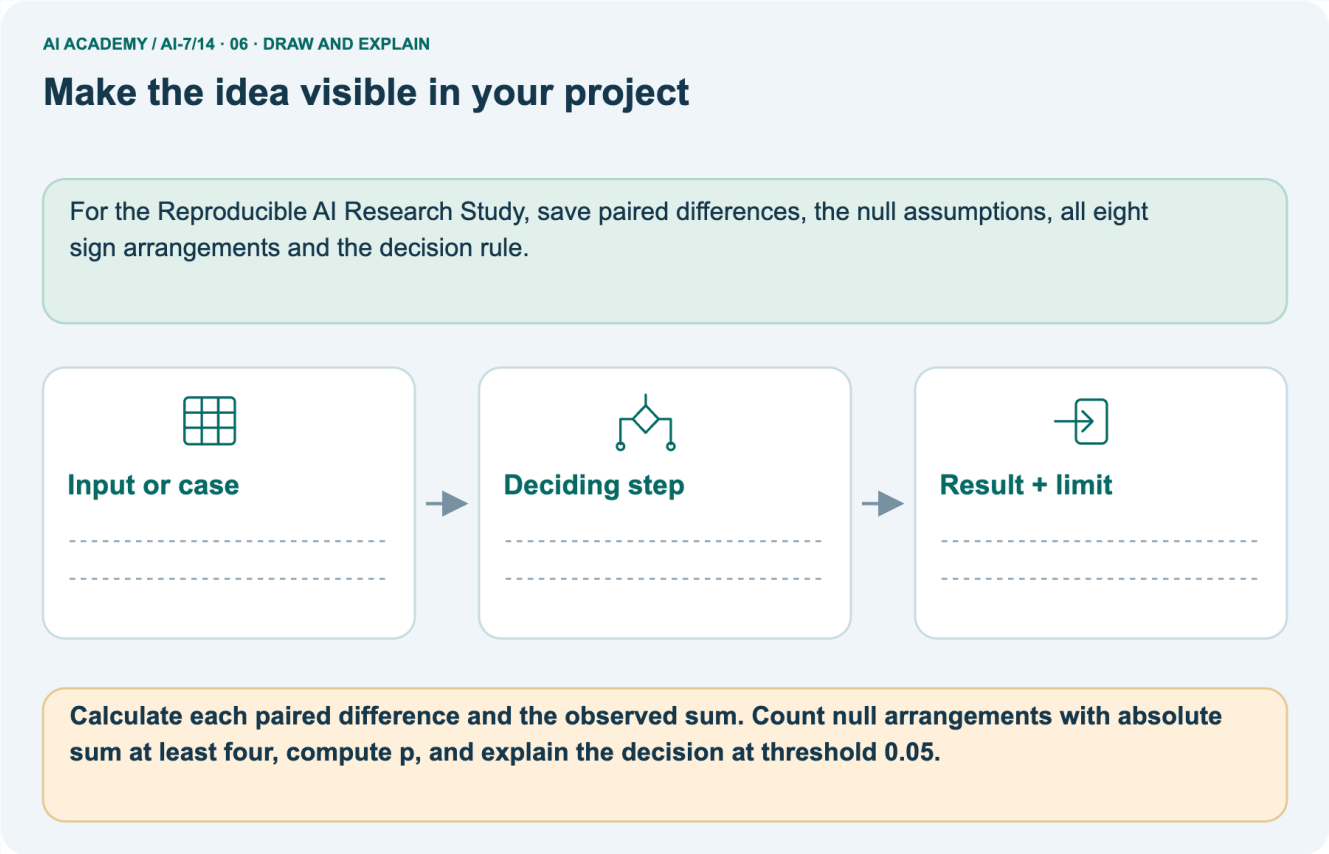
A test compares an observed statistic with outcomes under a specified **null model**. Here the sign-flip assumptions make eight arrangements equally likely.

At a predeclared 0.05 threshold, p=0.25 does not reject the null; it does not prove no difference. Limit: Three pairs provide coarse **evidence**.

The **p-value** is not the probability that the null is true; invalid independence or symmetry assumptions invalidate this test.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule. Draw your **input**, deciding step, result and limit.

Supported question: Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Statistical tests and assumptions: the case

02. Follow the deciding steps

03. Three independent paired task times

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.